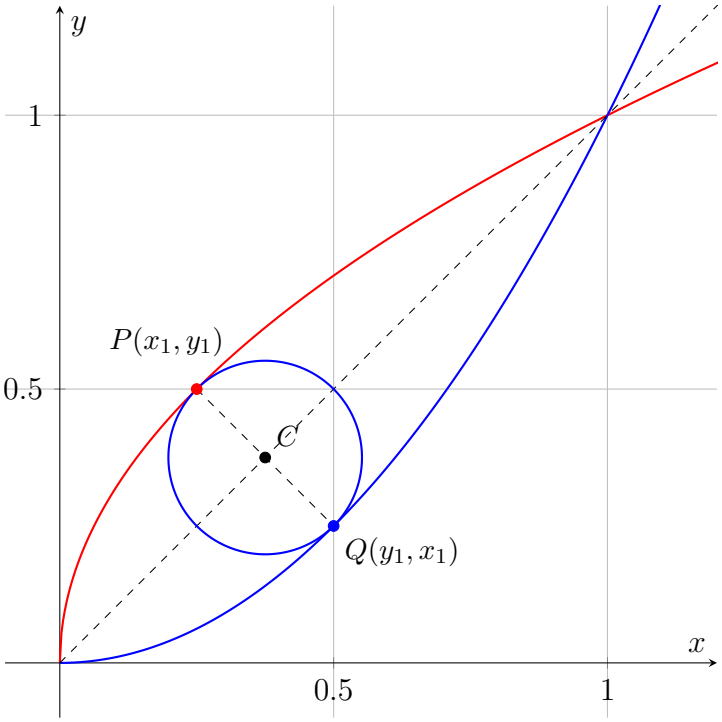
Ritangle 2026

Stage 1 - Question 10

Project π^*

October 17, 2026

Circle Geometry



*Solved by Nihal Sharma.

1 Problem

The problem statement was as follows:

Unknown

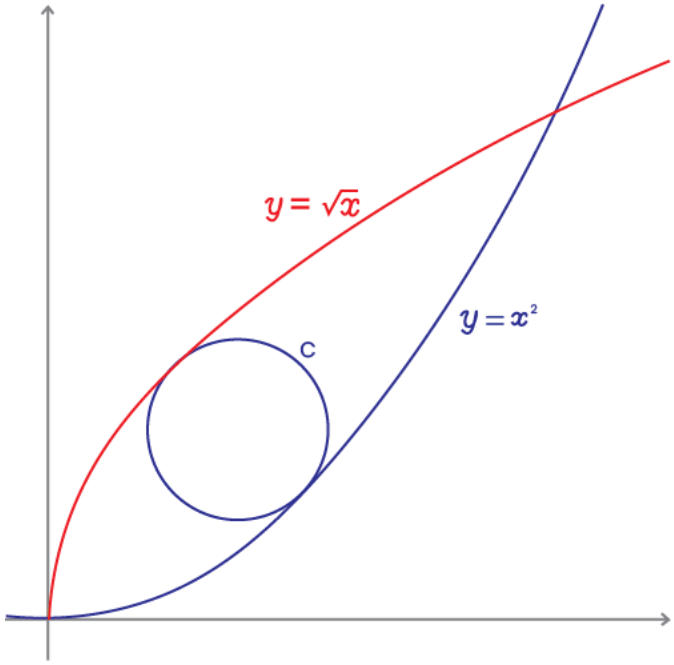
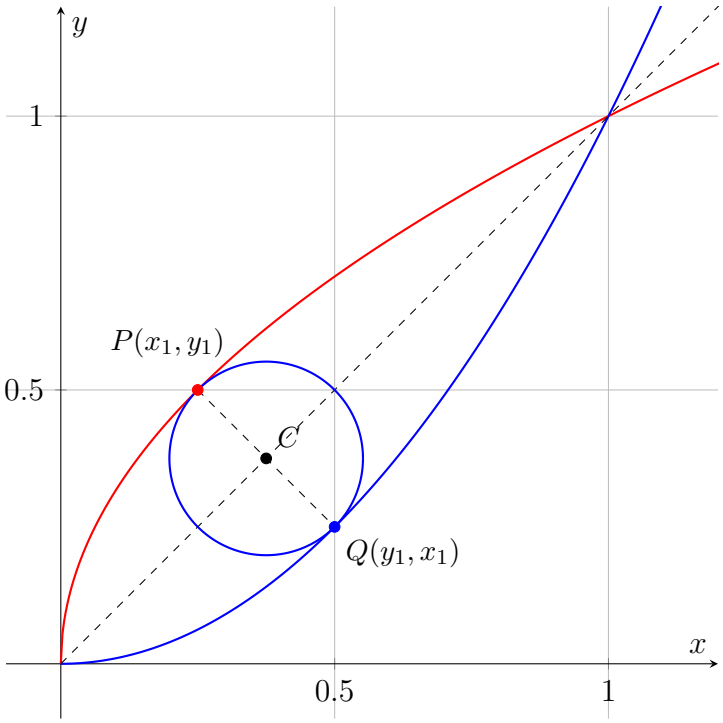


Figure 1: Reference image

I'm assuming the question is to calculate the area of the circle tangent to both curves

2 Solution

The key insight for this question is to recognise that the graph is symmetric about $y = x$



This is a rather useful, as from this we know that the point of tangency, $P = (x_1, y_1)$, when reflected gives the second point of tangency $Q = (y_1, x_1)$. This would imply the line segment PQ is a straight line passing through C. Additionally, this informs us C lies on $y = x$, so $C = (\alpha, \alpha)$ for some real α

From this we can conclude the following

$$\begin{aligned}
\text{gradient } PC &= \text{gradient } CQ \\
\frac{y_1 - \alpha}{x_1 - \alpha} &= \frac{x_1 - \alpha}{y_1 - \alpha} \\
\frac{(x_1 - \alpha)^2}{(y_1 - \alpha)^2} &= 1 \\
\therefore \frac{x_1 - \alpha}{y_1 - \alpha} &= \pm 1 \\
\therefore \frac{x_1 - \alpha}{y_1 - \alpha} &= -1 \because \text{gradient } < 0
\end{aligned}$$

Remark: The gradient of PQ can easily be deduced by realising it is formed by reflecting Q in the line $y = x$, thus must have gradient -1 (perpendicular lines). Nevertheless, the above is a good exercise of circle geometry fluency.

If the gradient of PQ is -1, then the gradient of the tangent at P and Q must be 1. By differentating $y = x^2$, $\frac{\mathrm{d}y}{\mathrm{d}x} = 2x = 1 \implies x = y_1 = 1/2 \therefore x_1 = 1/4$
From this we can deduce circle C has equation: $(x - 3/8)^2 + (y - 3/8)^2 = 1/32$

$\therefore$ the area of the circle is $\dfrac{\pi}{32}$

Final Answer Summary:

$$\text{Area} = \frac{\pi}{32} \quad \implies \quad \textit{Hint: ???}$$